



Lyntra Data

India's #1 Training Institute

Data Analytics Training

"1,00,000+ uplifted through our hybrid classroom & online training, enriched by real-time projects and job support."



3 Months



5 Live Projects



4.9/5

Lyntra Data Alumni Works @



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Fundamentals of IT & AI

Module 1: The Software Application Life Cycle

- What is an Application?
- Types of Applications
- Web Application Fundamentals
- Web Technologies:
 - **Frontend:** HTML, CSS, JavaScript, React
 - **Backend:** Python, Java, Node.js
 - **Databases:** SQL (MySQL, PostgreSQL), NoSQL (MongoDB)
- Software Development Life Cycle (SDLC)
 - **Phases:** Planning, Analysis, Design, Implementation (Coding), Testing, Deployment, Maintenance.
- Application Development Methodologies
 - **Agile:** Core principles, Scrum, Kanban
 - **Waterfall**

Module 2: Data Fundamentals

- What is Data?
- Types of Data
- Data Storage
- Data Analysis
- Data Engineering
- Data Science

Module 3: Computing and the Cloud

- The Importance of Computing Power
- Key Computing Technologies:
 - CPU (Central Processing Unit)
 - GPU (Graphics Processing Unit)
- Cloud Computing:
 - What is the Cloud?
 - Cloud Service Models:
 - IaaS (Infrastructure as a Service)
 - PaaS (Platform as a Service)
 - SaaS (Software as a Service)

Module 4: Introduction to AI and Generative AI

- What is Artificial Intelligence (AI)?
- How AI Works?
- Key Concepts:
 - Machine Learning (ML)
 - Deep Learning (DL)
- Generative AI:
 - What is Generative AI?
 - Examples: Large Language Models (LLMs), image generation models.

- AI in Everyday Learning

Module 5: Real-World Applications of Technology

- Customer Relationship Management (CRM)
- Human Resource Management Systems (HRMS)
- Retail & E-Commerce
- Healthcare

Basic Python

Module 1: Python Introduction and Setup

- Python's applicability across various domains
- Installation, environment setup, and path configuration
- Writing and executing the first Python script

Module 2: Python Fundamentals

- Keywords, Identifiers, and basic syntax
- Variables, Data Types, and Operators
- Introduction to Input/Output operations

Module 3: Control Structures and Functions

- Conditional Statements: If, Elif, Else
- Loops: For, While, and control flow mechanisms
- Understanding and defining Functions in Python

Module 4: Strings and Collections

- String operations and manipulations
- Lists and their operations

- Introduction to Tuples and Sets

Module 5: Diving Deeper into Collections

- Detailed exploration of Dictionaries
- Frozen Sets and their use-cases
- Advanced list comprehensions

Advanced Python

Module 1: Advanced Collections and Sequences

- Advanced methods in Lists, Tuples, and Dictionaries
- Sets, Frozen Sets, and operations
- Comprehensive look into Python Collections

Module 2: Functional Programming in Python

- Exploring types of Functions and Arguments
- Lambda functions and their applications
- Map, Reduce, and Filter functions

Module 3: File Handling and Modules

- File operations and handling different file formats
- Working with Excel and CSV files in Python
- Understanding and using Python Modules and Packages

Module 4: Object-Oriented Programming

- Deep dive into Classes, Objects, and Methods
- Constructors, Destructors, and Types of Methods
- Inheritance, Polymorphism, and Encapsulation

Module 5: Exception Handling and Regular Expressions

- Exception Handling: Try, Except, Finally
- Creating and using Custom Exceptions
- Utilizing Regular Expressions for pattern matching

SQL

Module 1: Introduction to Databases, RDBMS, and SQL Fundamentals

- **Database Concepts**
 - What is a Database? Understanding data storage and retrieval
 - RDBMS: Relational Database Management Systems and their advantages
 - Introduction to SQL: History, types, and use cases
- **SQL Server and SSMS**
 - Overview of SQL Server
 - Introduction to SQL Server Management Studio (SSMS)
 - Connecting, navigating, and exploring SSMS
- **Database Objects**
 - Tables, Views, Indexes, and Keys
 - Primary Key (PK) and Foreign Key (FK) concepts
- **Creating a Sample Database**
 - Steps to create a database and tables
 - Populating tables with sample data

Module 2: Data Query Language (DQL), SQL Operators, and Functions

- **Basic Queries**

- SELECT statement syntax and usage
- Filtering with WHERE clause
- Sorting with ORDER BY
- Using DISTINCT to remove duplicates

- **SQL Operators**

- Comparison operators: =, <>, >, <, BETWEEN, IN, LIKE
- Logical operators: AND, OR, NOT

- **SQL Functions**

- String functions: LEN, UPPER, LOWER
- Date functions: GETDATE, DATEPART, DATEADD
- Arithmetic operations on columns

- **Best Practices**

- Writing readable queries
- Commenting and structuring SQL code

Module 3: Joins, Aggregate Functions, and Grouping Techniques

- **Joins in SQL**

- INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN
- Self Join and Cross Join (Cartesian Product)
- Real-time examples and best practices

- **Aggregate Functions**

- COUNT, SUM, AVG, MIN, MAX
- Nested aggregations

- **Grouping Techniques**

- GROUP BY clause
- Filtering grouped data using HAVING vs WHERE
- Combining grouping with joins

Module 4: Advanced Queries, Subqueries, and Data Manipulation (DML & DDL)

- **Subqueries**

- Single-row and multi-row subqueries (IN, ANY, ALL)
- Correlated subqueries

- **Set Operators**

- UNION vs UNION ALL
- INTERSECT and EXCEPT

- **Data Manipulation Language (DML)**

- INSERT: Single and multiple rows
- UPDATE with conditions
- DELETE safely

- **Data Definition Language (DDL)**

- CREATE TABLE, ALTER TABLE, DROP TABLE
- Modifying table structures: adding, modifying, dropping columns
- Constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, DEFAULT
- Creating and dropping indexes

- **Best Practices**

- Safe modifications
- Version control and backups

Module 5: Data Control, Transactions, Stored Procedures, and Final Project

- **Data Control Language (DCL)**
 - GRANT and REVOKE: Managing user permissions
- **Transaction Control Language (TCL)**
 - BEGIN TRANSACTION, COMMIT, ROLLBACK, SAVEPOINT
 - Understanding ACID properties (Atomicity, Consistency, Isolation, Durability)
- **Stored Procedures**
 - What are stored procedures?
 - Creating, executing, and passing parameters
 - Benefits of using procedures
- **Views**
 - Creating, updating, and dropping views
 - Use cases for views in reporting and abstraction
- **Final Project**
 - Designing a small database
 - Applying DQL, DML, DDL, Joins, Aggregate functions, and Stored Procedures
 - Best practices for documentation and query optimization

Power BI

Module 1: Getting Started with Power BI

- Understanding Business Intelligence concepts
- Power BI components: Desktop, Service, Mobile
- Navigating the Power BI interface

- Connecting to data sources (Excel, CSV, databases, web)
- Exploring Power Query and creating your first report

Module 2: Data Preparation & Transformation

- Cleaning and shaping data in Power Query Editor
- Data profiling and handling missing/inconsistent values
- Transformations: merge, append, split, pivot, unpivot
- Creating calculated columns and custom tables
- Understanding relationships and data modeling basics

Module 3: Building Visual Reports

- Data visualization fundamentals and design principles
- Creating charts, tables, maps, and cards
- Adding slicers, filters, and interactions
- Using bookmarks, drill-downs, and drill-throughs
- Dashboard layout and storytelling best practices

Module 4: Introduction to DAX and Analytics

- What is DAX (Data Analysis Expressions)?
- Calculated columns vs measures
- Common DAX functions: SUMX, CALCULATE, FILTER, IF, etc.
- Creating KPIs, ratios, and time-based metrics
- Basic time intelligence (YTD, MTD, rolling averages)

Module 5: Sharing and Publishing Insights

- Publishing reports to Power BI Service
- Managing workspaces and dashboards online
- Setting up data refresh schedules

- Sharing reports securely with stakeholders
- Power BI ecosystem overview: integration with Excel, Teams, and SharePoint

Advanced Power BI

Module 1: Advanced Data Modeling & Query Optimization

- Star schema vs snowflake design
- Fact and dimension tables — best practices
- Modeling complex relationships and hierarchies
- Using calculation groups for advanced metrics
- Optimizing data models for performance and scalability

Module 2: Mastering DAX for Analytics

- Advanced filter context and row context
- Using variables and iterators effectively
- Time intelligence deep dive (QoQ, MoM, YOY comparisons)
- Dynamic measures and conditional calculations
- Performance tuning and debugging DAX queries

Module 3: Advanced Visualizations & Custom Dashboards

- Using Power BI visuals marketplace
- Integrating R and Python visuals for analytics
- Designing interactive dashboards for executives
- Dynamic visuals using parameters and field parameters
- Data storytelling with advanced formatting and themes

Module 4: Automation, Integration & AI Features

- Power BI and Power Automate integration

- AI visuals — Key Influencers, Decomposition Tree, Q&A
- Integrating with Azure ML and Cognitive Services
- Using APIs and embedding Power BI in applications
- Automating workflows with scheduled refresh and alerts

Module 5: Deployment, Governance & Enterprise BI

- Power BI Service administration and tenant settings
- Row-Level and Object-Level Security (RLS/OLS)
- Managing large datasets and dataflows
- Deployment pipelines and version control
- Governance, monitoring, and best practices for enterprise BI

Pandas

Module 1: Introduction to Pandas

- Overview and core concepts of Pandas
- Understanding Series and DataFrames
- Basic operations, attributes, and methods
- Comparison of Pandas with NumPy

Module 2: Data Importing & Exporting

- Reading data from CSV, Excel, JSON, and SQL files

- Writing data to different file formats
- Handling file paths and directories
- Managing large datasets efficiently

Module 3: Data Selection, Cleaning & Manipulation

- Indexing and subsetting data
- Conditional selection and querying
- Handling missing values, duplicates, and outliers
- Sorting and ranking data
- Data transformations using `apply()`, `map()`, and `replace()`

Module 4: Data Aggregation & Combination

- Grouping data using `groupby()`
- Aggregation and transformation operations
- Combining datasets through merging, joining, and concatenation
- Summarizing and restructuring data

Module 5: Advanced Pandas Operations

- Pivot tables and cross-tabulations
- Working with MultiIndex and hierarchical data
- Time series analysis and date-time operations
- Window functions (`rolling`, `expanding`)
- Performance optimization and best practices

Numpy

Module 1: Introduction to NumPy

- Overview and importance of NumPy
- Creating and initializing arrays

- Basic array operations and mathematical functions
- Differences between Python lists and NumPy arrays

Module 2: Data Generation with NumPy

- Generating arrays using `arange()`, `linspace()`, and `ones()`, `zeros()`, `eye()`
- Creating arrays from existing data (lists, tuples)
- Loading and saving data with NumPy
- Data type specification and conversions

Module 3: Working with Array Attributes & Indexing

- Understanding array attributes (`shape`, `size`, `ndim`, `dtype`, etc.)
- Indexing and slicing arrays
- Boolean and fancy indexing
- Iterating over arrays efficiently

Module 4: Array Manipulations & Linear Algebra

- Reshaping, stacking, splitting, and broadcasting arrays
- Transposing and flattening arrays
- Matrix operations and dot products
- Solving linear equations using NumPy's `linalg` module

Module 5: Random Number Generation & ML Utilities

- Generating random numbers using `numpy.random`
- Sampling, shuffling, and seeding
- Statistical operations (mean, median, variance, std)
- NumPy utilities for machine learning workflows (normalization, vectorization, etc.)

Tools



AI-powered coding assistant for efficiency and automation.



Git is a distributed version control system for tracking code changes.



Jenkins is an open-source tool for automating CI/CD pipelines.



Kubernetes automates deployment, scaling, and management of containers.



AI model for generating and assisting with code and text.



Google's AI model for assisting in code generation, debugging, and research.



Power BI is a business analytics tool for visualizing data insights.



Version control tools for managing and collaborating on test automation scripts.



Python powers everything from web apps to AI.



Core framework for building autonomous AI agents and workflows.

Tools



Linux is an open-source operating system for servers, desktops, and devices.



Nexus is a repository manager for managing build artifacts and dependencies.



Slack is a messaging platform for teams and workplace collaboration.



Trello is a visual tool for organizing tasks and projects.



SQL is a language for managing and querying relational databases.



Lightweight and customizable code editor with Python support.



Interactive coding environment for testing DSA concepts.



One of the best platforms for solving DSA problems.



Step-by-step visualization of code execution.

